Appendix B

Nutritional and functional roles of minerals in foods

Mineral	Source	Function
Aluminium	Low and variable in foods	Possibly essential, evidence not conclusive. Deficiency unknown. Leavening agent: as sodium aluminum sulfate ($Na_2SO_4 \cdot Al_2(SO_4)_3$). Texture modifier.
Bromine	Brominated flour	Not known to be essential to humans. Dough improver: $KBrO_3$ improves baking quality of wheat flour.
Calcium	Dairy products, green leafy vegetables, tofu, fish bones	Essential nutrient: deficiency leads to osteoporosis in later life. Texture modifier: forms gels with negatively charged macromolecules such as alginates, low-methoxyl pectins, soy proteins, caseins, etc. Firms canned vegetables when added to canning brine.
Copper	Organ meats, seafood, nuts, seeds	Essential nutrient: deficiency rare. Catalyst: lipid peroxidation, ascorbic acid oxidation, non-enzymatic oxidative browning. Colour modifier: may cause black discoloration in canned, cured meats. Enzyme cofactor: polyphenoloxidase. Texture stabiliser: stabilises egg-white foams.
Iodine	Iodised salt, seafood, plants and meat from animals grown in areas where soil iodine is not depleted	Essential nutrient: deficiency produces goitre and cretinism. Dough improver: KIO_3 improves baking quality of wheat flour.

Iron	Cereals, legumes, meat, contamination from iron utensils and soil, enriched foods	Essential nutrient: deficiency leads to anaemia, impaired immune response, reduced productivity, impaired cognitive development in children. Excessive iron stores may increase risk of cancer and heart disease. Catalyst: Fe^{2+} and Fe^{3+} catalyse lipid peroxidation in foods. Colour modifier. Colour of fresh meat depends on valence of Fe in myoglobin and hemoglobin: Fe^{2+} is red, Fe^{3+} is brown. Forms green, blue or black complexes with polyphenolic compounds. Reacts with S^{2-} to form black FeS in canned foods. Enzyme cofactor: lipoxygenase, cytochromes, ribonucleotide reductase, etc.
Magnesium	Whole grains, nuts, legumes, green leafy vegetables	Essential nutrient: deficiency rare. Colour modifier: removal of Mg from chlorophyll changes colour from green to olive-brown.
Manganese	Whole grains, fruits, vegetables	Essential nutrient: deficiency extremely rare. Enzyme cofactor: pyruvate carboxylase, superoxide dismutase.
Nickel	Plant foods	Essential nutrient: deficiency in humans unknown. Catalyst: hydrogenation of vegetable oils – finely divided, elemental Ni is the most widely used catalyst for this process.
Phosphates	Ubiquitous	Essential nutrient: deficiency rare due to presence in virtually all foods. Acidulent: H_3PO_4 in soft drinks. Leavening acid: $Ca(HPO_4)_2$ is a fast-acting leavening acid. Moisture retention in meats: sodium tripolyphosphate improves moisture retention in cured meats. Emulsification aid: phosphates are used to aid emulsification in comminuted meats and in processed cheeses.
Potassium	Fruits, vegetables, meats	Essential nutrient: deficiency rare. Salt substitute: KCl may be used as a salt substitute. May cause bitter flavour. Leavening agent: potassium acid tartrate.
Selenium	Seafood, organ meats, cereals (depending on levels in soil)	Essential nutrient: Keshan disease (endemic cardiomyopathy in China) was associated with selenium deficiency. Low selenium status may be associated with increased risk for cancer and heart disease. Enzyme cofactor: glutathione peroxidase.
Sodium	Food additives, milk, low in most raw foods	Essential nutrient: deficiency is rare; excessive intakes may lead to hypertension. Flavour modifier: NaCl elicits the classic salty taste in foods. Preservative: NaCl may be used to lower water activity in foods. Leavening agents: many leavening agents are sodium salts, e.g. sodium bicarbonate, sodium aluminium sulphate, sodium acid pyrophosphate.
Sulphur	Widely distributed	Essential nutrient: a constituent of the essential amino acids methionine and cystine. Sulphur amino acids

		may be limiting in some diets. Browning inhibitor: sulphur dioxide and sulphites inhibit both enzymatic and non-enzymatic browning. Widely used in dried fruits. Anti-microbial: prevents, controls microbial growth. Widely used in wine making.
Zinc	Meats, cereals	Essential nutrient: deficiency produces loss of appetite, growth retardation, skin changes. Marginal deficiency exists in US and Europe but extent is unknown. Pronounced deficiency was documented in populations in the Middle East. ZnO is used in the lining of cans for protein rich foods to lessen formation of black FeS during heating. Zn can be added to green beans to help stabilise the colour during canning.

From Miller, D.R. (1996) 'Minerals'. In: O.R. Fennema (ed.) *Food Chemistry*. Marcel Dekker, pp. 617–47.